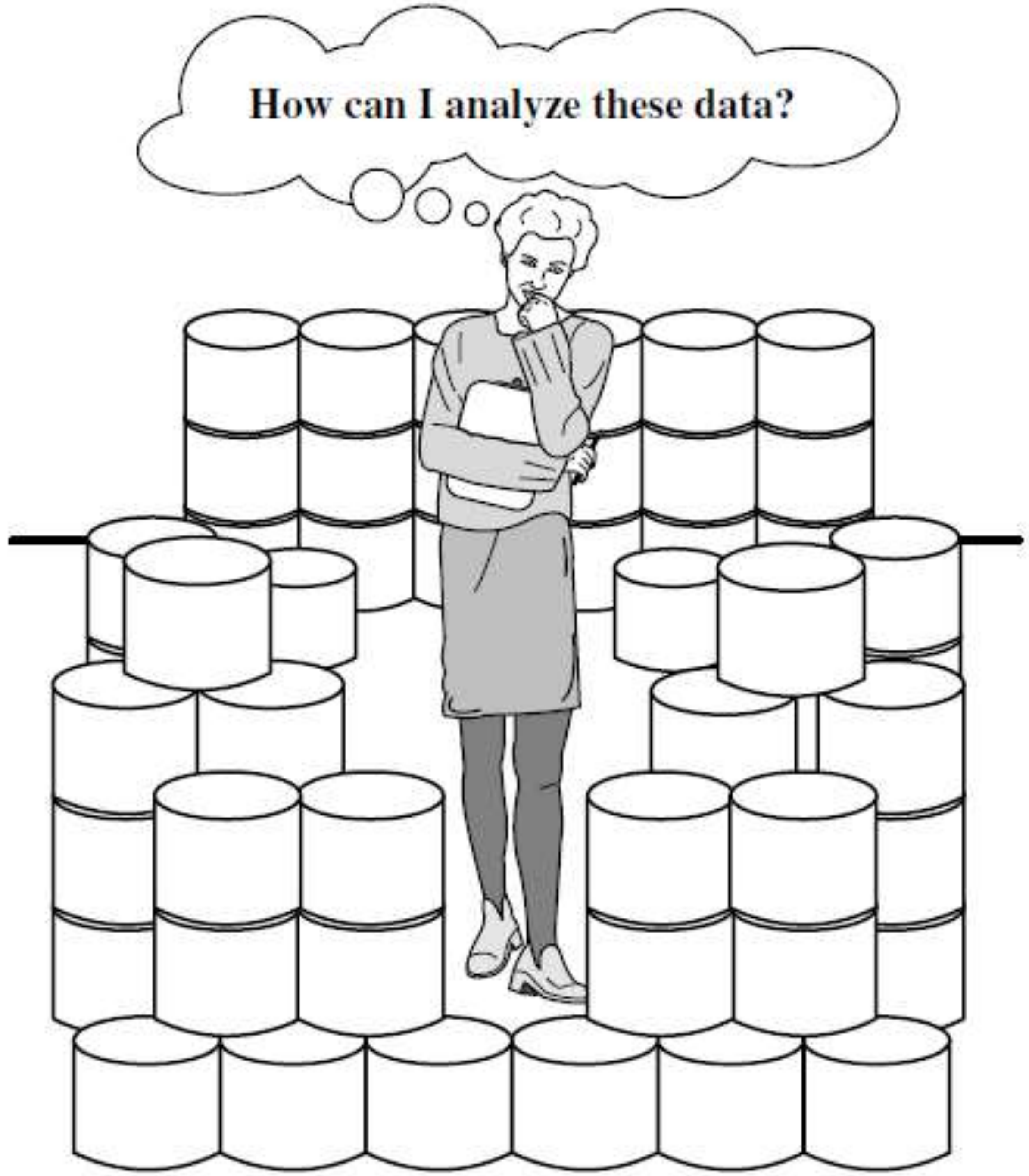
Data Mining

Data mining

Unit I

Introduction: Introduction to Data Mining-Types of Data and Patterns Mined- Technologies- Applications-Major Issues in Data Mining

Why Data mining ?

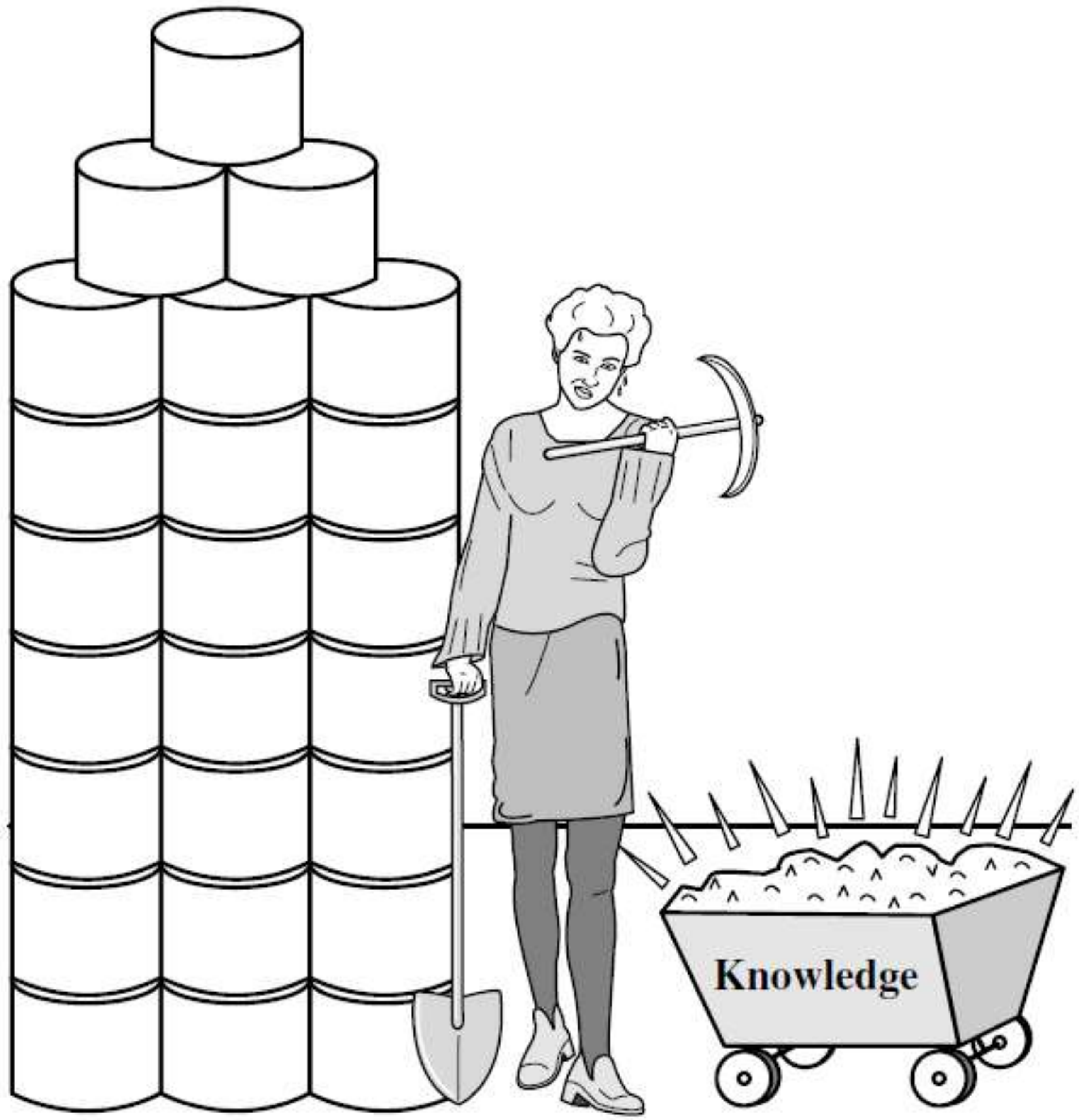


The world is Data rich but information is very poor

Why Data mining ?

- Rapid growth of data from multiple sources.
- Major Sources of Abundant Datas
  - Business: Web, e-commerce, transactions, stocks
  - Science: Remote sensing, bioinformatics, simulations
  - Society: News, digital cameras, YouTube, social media
- Need to extract useful knowledge and insights
- It helps in decision making and predictive analysis
- Widely used in various fields: finance, healthcare, marketing….

What is Data mining ?



Data mining: Searching for Knowledge (interesting patterns) in data

What is Data mining ?

- Extracting patterns or knowledge from large datasets
- Also known as **Knowledge Discovery in Databases (KDD)**
- Involves techniques from machine learning, statistics, and database systems
- Goal: Turn raw data into useful information

Knowledge Discovery from Data(KDD)

- Understanding the Domain: Identify prior knowledge and define application goals
- Data Preparation : Create target dataset (data selection)
- Data cleaning & preprocessing
- Data Reduction & Transformation : Identify and retain useful features

    Apply: Dimensionality/variable reduction, Invariant representation
- Select Functions of Data Mining
    - Summarization
    - Classification
    - Regression
    - Association
    - Clustering

Knowledge Discovery from Data(KDD)

- Choosing the Mining Algorithm(s) :Select the most suitable algorithm for the task
    - Decision trees , Neural networks , Clustering methods , Association rule mining, etc.
- Data Mining Actual mining step: Extract patterns of interest from the dataset, Applies algorithms to identify useful trends or rules
- Pattern Evaluation & Knowledge Presentation :
    - Evaluate discovered patterns for: Usefulness, Interestingness, Redundancy removal
    - Present knowledge using: Visualization tools, Transformation technique
- Use of Discovered Knowledge :
    - Apply the knowledge to: Business , decisions, Scientific discovery, Process optimization, Policy making, etc

The knowledge discovery process

1. **Data cleaning :** to remove noise and inconsistent data

2. **Data integration :** where multiple data sources may be combined

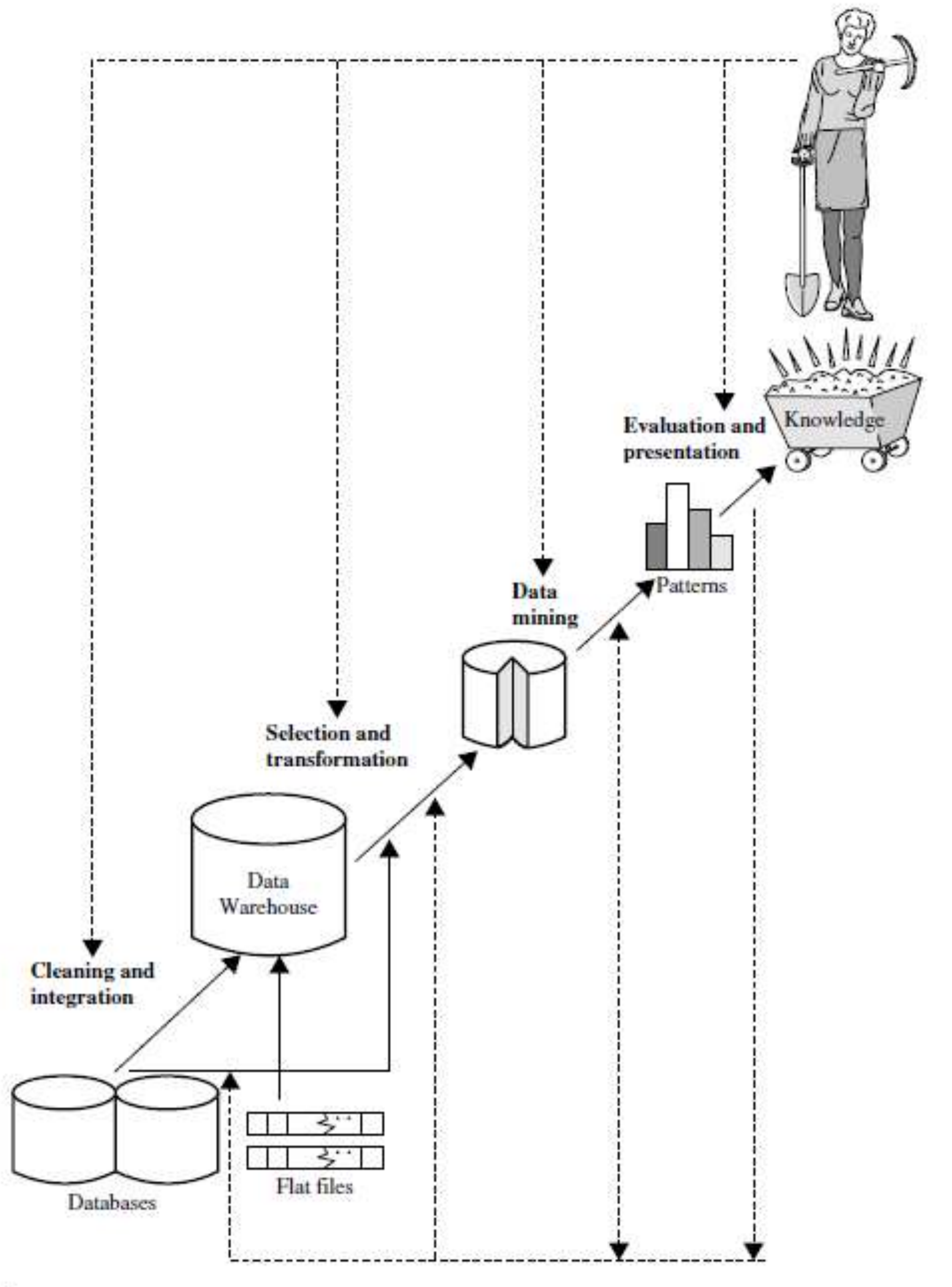
3.**Data selection :**where data relevant to the analysis task are retrieved from the database

4. **Data transformation :**where data are transformed and consolidated into forms appropriate for mining by performing summary or aggregation operations.

5**. Data mining :** an essential process where intelligent methods are applied to extract data patterns

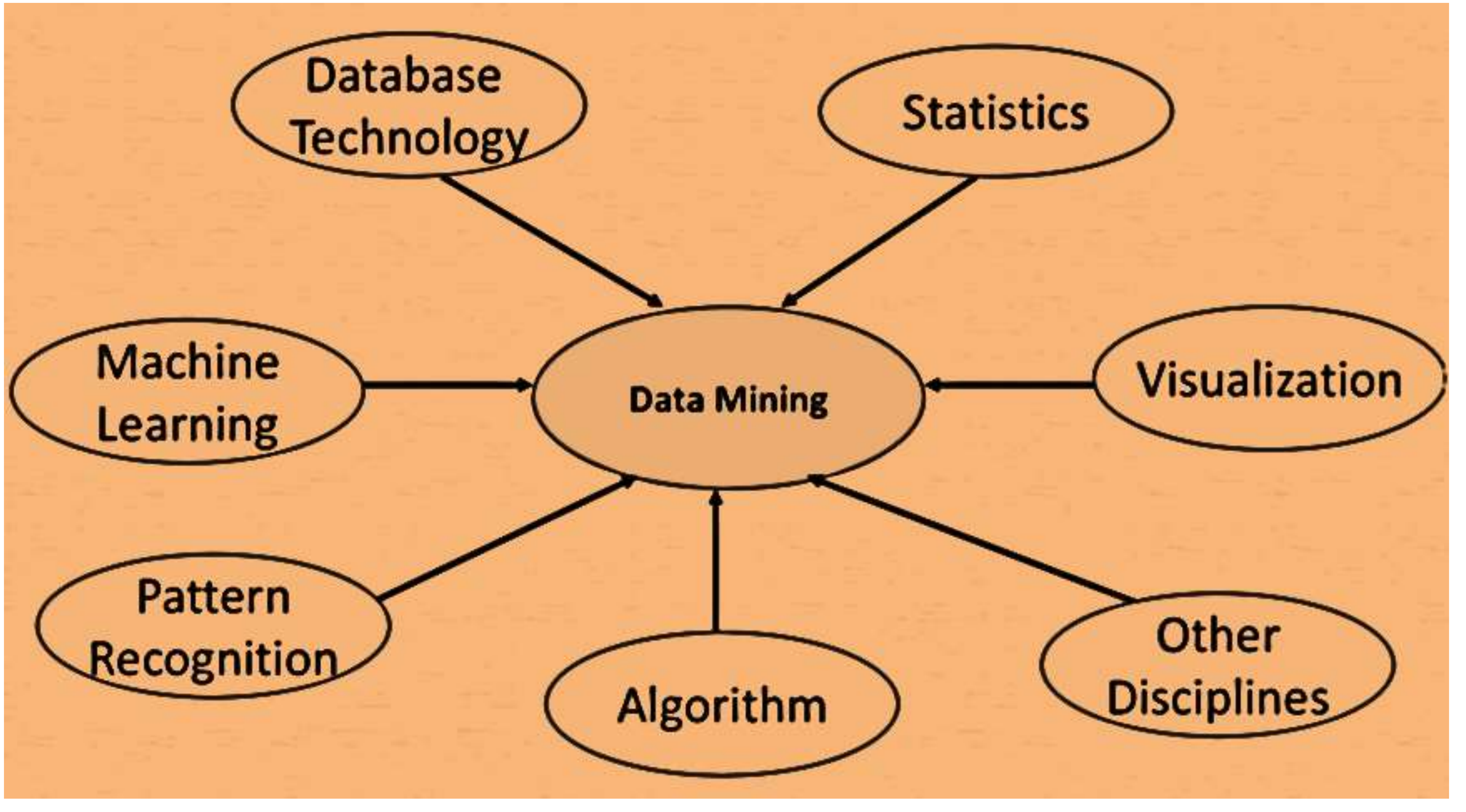
6**. Pattern evaluation :** to identify the truly interesting patterns representing knowledge based on interestingness measures.

7. **Knowledge presentation**



Data mining Steps in the process of Knowledge discovery

Data Mining – A Confluence of Multiple Disciplines



Data mining on What kind of data

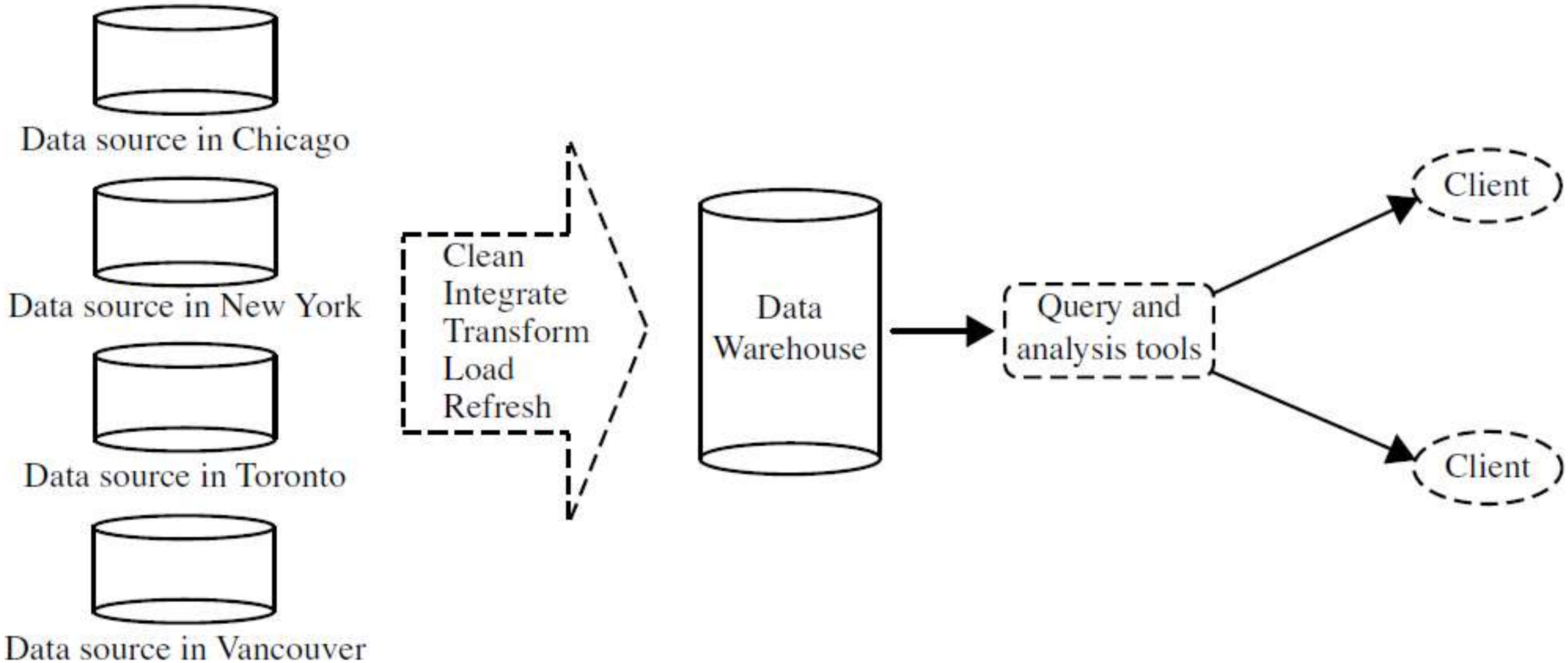
- Database data
- Data warehouse data
- Transactional data
- Other kinds of data (e.g., data streams, ordered/sequence data,graph or networked data, spatial data, text data,multimedia data, and WWW)

Database data

- Database management system (DBMS), consists of a collection of interrelated data, known as a database, and a set of software programs to manage and access the data
- A relational database is a collection of tables, each of which is assigned a unique name.
- Each table consists of a set of attributes (columns or fields) and usually stores a large set of tuples (records or rows).
- Relational data can be accessed by database queries written in a relational query language (e.g., SQL) or with the assistance of graphical user interfaces

DataWarehouses

- A data warehouse is a repository of information collected from multiple sources, stored under a unified schema.
- It usually residing at a single site.
- Data warehouses are constructed via a process of
  - Data cleaning
  - Data integration
  - Data transformation
  - Data loading
  - Periodic data refreshing.



Framework for a Data warehouse

Transactional Data

- Transactional database captures a transaction, such as a
  - customer’s purchase,
  - a flight booking, or a user’s clicks on a web page.
- A transaction typically includes
  - a unique transaction identity number (trans ID) and a list of the items making up the transaction, such as the items purchased in the transaction.
- A transactional database may have additional tables, which contain other information related to the transactions, such as
  - item description
  - information about the salesperson or the branch.

trans_ID	*list_of_item_IDs*
T100	I1, I3, I8, I16
T200	I2, I8
...	...

A fragment of transactional data

Other Kinds of Data

- Time-related or sequence data (e.g., historical records, stock exchange data, and time-series and biological sequence data)
- Data streams (e.g., video surveillance and sensor data, which are continuously transmitted)
- Spatial data (e.g., maps), engineering design data (e.g., the design of buildings, system components, or integrated circuits)
- Hypertext and multimedia data (including text, image, video, and audio data)
- Graph and networked data (e.g., social and information networks),
- Web (a huge, widely distributed information repository made available by the Internet).

Data mining types:

- Descriptive mining

 Descriptive mining tasks characterize properties of the data in a target data set.

- Predictive mining

 Predictive mining tasks perform induction on the current data in order to make predictions.

Types of patterns

1. Frequent Patterns:

Patterns (like itemset, subsequences, or substructures) that appear frequently in a dataset.

Example: In a supermarket:{Milk, Bread} → {Butter} (i.e., people who buy milk and bread often buy butter).

2. Sequential Patterns;

Patterns that capture sequences of events or actions.

Example: A customer first buys a phone → then a phone case → then a power bank.

3. Association Patterns:

Discover relationships between variables using rules like:Support and Confidence,

Example:Diapers → water(unexpected but often found in market basket data).

4. Correlation Patterns:

Identify the statistical relationships between variables.

Strong correlations indicate that variables move together.

Example: High income is positively correlated with luxury car purchases.

5. Clusters (Clustering Patterns):

Natural groupings of data items based on similarity.

No predefined labels.

Example: Customer segmentation (grouping similar customers for marketing).

6. Classification Patterns:

Patterns that assign data into predefined categories.

Often learned from labeled training data.

Example: Email classification as spam or not spam.

7. Anomaly or Outlier Patterns:

Identify rare or unusual data points that do not conform to expected patterns.

Example: Credit card fraud detection.

8. Trend Patterns:

Patterns that show a direction of change over time.

Example: Increasing sales over months

Major Issues in Data Mining

1. Data Quality

Quality of data is paramount in **data mining**. Issues such as **noisy data**, **incomplete data**, and **unstructured data** can significantly impair the effectiveness of genuine data mining measure and efforts. Data must be accurate, complete, and consistent for **genuine data mining measures** to succeed.

Solutions:

- Implement robust **data cleaning methods** to preprocess data and improve quality.
- Use **data mining tools** that can handle **heterogeneous data sources** and **diverse data types**.
- Continuously monitor and update data quality standards.

2. Data Privacy and Security

As data mining often involves analyzing sensitive information, ensuring **data privacy and security** is crucial. Unauthorized access to data can lead to privacy and data security breaches and misuse of personal information.

Solutions:

•Employ encryption and access controls to protect data.

•Develop policies and frameworks that adhere to data privacy regulations.

•Utilize **data mining techniques** that minimize privacy risks, such as anonymization and differential privacy.

3. Handling Complex and Diverse Data Types

•Data mining is often challenged by the need to process complex types of data such as spatial data, temporal data, and media data. These data types require specialized approaches to analyze and extract useful insights.

Solutions:

•Leverage complex data perception methods tailored to specific data types.

•Use parallel data mining algorithms and distributed data mining algorithms to handle large-scale and complex datasets.

•Develop flexible data mining tools that can adapt to various data formats and structures.

4.Scalability and Efficiency

With the growth of data volume, scalability and efficiency refining data mining requests become critical issues. Data miners must be able to handle **enormous data sets** without compromising performance.

Solutions:

•Utilize **distributed data mining algorithms** that distribute tasks across multiple systems.

•Optimize **mining algorithms** to improve their performance on large datasets.

•Invest in high-performance computing resources to support intensive data mining tasks.

5. Integration with Heterogeneous Data Sources

Data often comes from multiple sources with varying formats, making integration into data warehouses a significant challenge. Combining data from different systems requires careful handling to ensure consistency and reliability.

Solutions:

•Use **data integration tools** to combine data from different sources into a **unified data archive**.

•Implement ETL (Extract, Transform, Load) processes to standardize data from heterogeneous sources.

•Develop metadata systems to manage and align data across different platforms.

6. Interpretation and Usability of Results

The ultimate goal of a data mining algorithm is to provide actionable insights. However, interpreting the **discovered patterns** and ensuring their usability can be difficult, especially for non-technical stakeholders.

Solutions:

•Enhance **data visualization** capabilities to make results more accessible and understandable.

•Provide training and support for users to interpret and act on data mining insights.

•Develop interactive systems that allow users to explore and refine data mining outputs.

8.Legal and Ethical Concerns

Data mining can raise significant legal and ethical issues, particularly concerning the use of personal data. Ensuring compliance with regulations and addressing ethical considerations are crucial.

Solutions:

•Adhere to legal standards such as GDPR and CCPA to ensure data handling complies with regulations.

•Develop ethical guidelines and practices for data mining to safeguard individual rights.

•Engage stakeholders in discussions about the ethical use of data and the impact of data mining practices.

Applications of DataMining

Business

- Predicting the bankruptcy of a business firm
- Prediction of bank loan defaulters
- Prediction of interest rates for corporate funds and treasury bills
- Identification of groups of insurance policy holders with average claim cost

Applications of DataMining

Telecommunication

- Trend analysis and Identification of patterns to diagnose chronic faults
- To detect frequently occurring alarm episodes and its prediction
- To detect bogus calls, fraudulent calls and identification of its callers
- To predict cellular cloning fraud.

Applications of DataMining

Marketing

- Retail sales analysis
- Market basket analysis
- Product performance analysis
- Market segmentation analysis
- Analysis of mail depth to identify customers who respond to mail campaigns.
- Study of travel patterns of customers.

Applications of DataMining

Medicine

- Prediction of diseases given disease symptoms
- Prediction of effectiveness of the treatment using patient history

Applications of DataMining

Security

- Face recognition/Identification
- Biometric projects like identification of a person from a large image or video database.
- Applications involving multimedia retrieval are also very popular.